

team-dockerlocker

Midterm Evaluation: DockerLocker (ism6562-midterm-DockerLocker)

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Repository: <https://github.com/hrashid13/ism6562-midterm-DockerLocker> (upstream) **Level:** Graduate (ISM 6562) **Presentation Topic:** Apache Ignite

Part 1: Operational Databases, Data Warehousing, and ETL (20 points)

Section 1: Environment Setup (3 pts)

Task	Points	Score	Notes
1.1 Docker compose ps	1	1	Documented in Word doc
1.2 \dt for all databases	1	1	Present
1.3 pgAdmin screenshot	1	1	Present

Subtotal: 3/3

Section 2: Explore Operational Databases (4 pts)

Task	Points	Score	Notes
2.1 Sales schema + normalization	1.5	1.5	Present
2.2 HR schema + relationships	1.5	1.5	Present
2.3 Four queries	1	1	Present

Subtotal: 4/4

Section 3: Data Warehouse Concepts (4 pts)

Task	Points	Score	Notes
3.1 Star schema tables	0.5	0.5	Present
3.2 Operational vs warehouse comparison	2	2	Addresses all areas
3.3 Facts vs dimensions	1.5	1.5	Correct

Subtotal: 4/4

Section 4: ETL Pipeline (5 pts)

Task	Points	Score	Notes
4.1 ETL script analysis	2	2	All phases covered
4.2 Count verification	1	1	Counts present
4.3 New order through ETL	2	2	Flow documented

Subtotal: 5/5

Section 5: Analytical Queries (4 pts)

Task	Points	Score	Notes
5.1 Three analytical queries	3	3	SQL in Part_5_Queries.sql; correct queries
5.2 EXPLAIN ANALYZE comparison	1	1	Present with comparison

Subtotal: 4/4

Part 1 Total: 20/20

Part 2: Sharding the Sales Database (20 points)

Section 1: Setup & Exploration (4 pts)

Task	Points	Score	Notes
1.1 Docker compose ps	1	1	Six services shown
1.2 Count rows	1	1	Correct counts
1.3 Uneven distribution analysis	2	2	Hot shard analyzed

Subtotal: 4/4

Section 2: Update ETL Pipeline (8 pts)

Task	Points	Score	Notes
2.1 Diagnose ETL failure	2	2	Correct diagnosis
2.2 Modified ETL script + output	4	4	Full working ETL (etl.py and etl_script.py both present with identical content); correctly extracts from both shards, merges results
2.3 Verify warehouse data + new order	2	2	Verified

Subtotal: 8/8

Section 3: Cross-Shard Challenges (4 pts)

Task	Points	Score	Notes
3.1 Cross-shard query + limitation	2	2	Both outputs documented
3.2 Sharding trade-offs	2	2	Comprehensive analysis

Subtotal: 4/4

Section 4: EXPLAIN ANALYZE (4 pts)

Task	Points	Score	Notes
4.1 SE shard EXPLAIN	2	2	Output shown
4.2 Warehouse EXPLAIN + response	2	2	Isolation explained

Subtotal: 4/4

Part 2 Total: 20/20

Part 3: Adding Cassandra for Real-Time Order Ingestion (20 points)

Section 1: Setup & Cassandra Exploration (4 pts)

Task	Points	Score	Notes
1.1 Docker compose ps	1	1	Eight services documented
1.2 Nodetool + DESCRIBE	1	1	Cluster and schema shown
1.3 Table design comparison	2	2	Good comparison

Subtotal: 4/4

Section 2: Query-First Design Analysis (4 pts)

Task	Points	Score	Notes
2.1 Valid Cassandra queries	2	2	Outputs present
2.2 Invalid query + explanation	2	2	Error and explanation present

Subtotal: 4/4

Section 3: Write Orders to Cassandra (4 pts)

Task	Points	Score	Notes
3.1 Dual inserts + verification	2	2	Present
3.2 Dual-write consistency	2	2	Present

Subtotal: 4/4

Section 4: Update ETL Pipeline (4 pts)

Task	Points	Score	Notes
4.1 ETL analysis	2	2	Changes identified
4.2 Modified ETL + verification	2	2	Cassandra extraction working; warehouse verified

Subtotal: 4/4

Section 5: Polyglot Persistence Reflection (4 pts)

Task	Points	Score	Notes
5.1 Comparison table + response	2	2	Present and accurate
5.2 Cassandra as warehouse argument	2	2	Strong argument against

Subtotal: 4/4

Part 3 Total: 20/20

Part 4: Distributed SQL Database Presentation (40 points)

Database Selected: Apache Ignite

Note on scoring: Full marks awarded for the in-class presentation delivery portion (36 points: Technical Accuracy, TechRetail Evaluation, Presentation Quality, Visual Aids, Q&A Handling, Equal Participation). While there was room for improvement on each presentation, the feedback given and discussions held during class provide overall guidance for areas of growth. Full marks reflect the lengthy delay between the presentations and these evaluations being returned.

Category	Points	Score	Notes
Technical Accuracy	8	8	Watch for: Ignite is NOT "just a cache" -- it's a full distributed SQL database with disk persistence. Uses Raft for partition replication. Memory-first architecture with WAL.
TechRetail Evaluation	8	8	Ignite could replace sales shards for low-latency operations. Co-located compute is a key differentiator.
Presentation Quality	8	8	Presentation is hosted on OneDrive (external link in repo).
Visual Aids	6	6	Full marks.
Q&A Handling	2	2	Full marks.
Equal Participation	4	4	Full marks.
Slide Content (/10)	—	0	No PPTX submitted to repo. Only a text file with a OneDrive link (Part 4/Group Powerpoint). Cannot evaluate slide content. The rubric requires the slide deck to be submitted; an external link is not a deliverable.

Category	Points	Score	Notes
GitHub Repository	4	4	Outstanding. Repo organized with Part 1-4 directories plus a comprehensive demo/ directory containing a working 3-node Apache Ignite cluster with 553k rows of grocery store data. All 5 members have commits (Hesham: 7, Freeman: 3, Gustavo: 2, Zachary: 1, Savannah: 1). Prof-tcsmith has access. Full Docker infrastructure for parts 2-3. Demo includes distributed SQL queries, fault tolerance testing, and transaction isolation level demonstrations.

Part 4 Total: 40/40

GitHub Repository Assessment

Criterion	Status
Repo organized by parts	Yes -- Part 1/ through Part 4/
Prof-tcsmith invited	Yes
All members have commits	Yes -- all 5 members
Part 4 materials in repo	OneDrive link in Part 4/; full demo in demo/
Submission docs present	Yes -- .docx for parts 1 and 3; .sql for part 1
Modified ETL scripts	Yes -- Part 2 and Part 3 etl/ directories
Bonus: Working demo	demo/ contains a full 3-node Apache Ignite cluster with schema, 553k rows of data, 9 demo queries, fault tolerance testing, and transaction isolation testing. Includes Docker Compose, Python load scripts, and comprehensive README. This is exceptional extra work.

Summary

Component	Points Available	Score
Part 1 -- Operational Databases & ETL	20	20
Part 2 -- Sharding	20	20
Part 3 -- Cassandra	20	20
Part 4 -- Repository	4	4
Part 4 -- Presentation delivery (in-class)	36	36
Total	100	100/100

Strengths

- **Exceptional demo work** -- the 3-node Apache Ignite cluster with 553k rows of grocery data, 9 demo queries, fault tolerance testing, and transaction isolation demos goes far beyond what was required. This demonstrates genuine engagement with distributed database concepts.

- Solid technical execution across all three hands-on parts
- Full Docker infrastructure with health checks, proper networking, and volume management
- Savannah's SQL transaction/consistency testing in the demo shows advanced understanding
- Good PR-based workflow evident in git history

Areas for Improvement

- Part 4 presentation is an external OneDrive link rather than committed to the repo
- Gustavo, Zachary, and Savannah each have only 1-2 commits -- contribution could be more balanced
- Some duplicate files (etl.py and etl_script.py are identical in Part 2)